

EMT:

→ Vector addition and subtraction:

$$\begin{aligned} \vec{U} &= U_x \hat{a}_x + U_y \hat{a}_y + U_z \hat{a}_z \\ \vec{V} &= V_x \hat{a}_x + V_y \hat{a}_y + V_z \hat{a}_z \end{aligned} \quad \left. \right\} \text{Cartesian Coordinates } [x, y, z] \quad \begin{cases} -\infty \leq x \leq \infty \\ -\infty \leq y \leq \infty \\ -\infty \leq z \leq \infty \end{cases}$$

$$\begin{aligned} \vec{U} &= U_\rho \hat{a}_\rho + U_\phi \hat{a}_\phi + U_z \hat{a}_z \\ \vec{V} &= V_\rho \hat{a}_\rho + V_\phi \hat{a}_\phi + V_z \hat{a}_z \end{aligned} \quad \left. \right\} \text{Cylindrical Coordinates } [\rho, \phi, z] \quad \begin{cases} 0 \leq \rho \leq \infty \\ 0 \leq \phi \leq 2\pi \\ -\infty \leq z \leq \infty \end{cases}$$

$$\begin{aligned} \vec{U} &= U_r \hat{a}_r + U_\theta \hat{a}_\theta + U_\phi \hat{a}_\phi \\ \vec{V} &= V_r \hat{a}_r + V_\theta \hat{a}_\theta + V_\phi \hat{a}_\phi \end{aligned} \quad \left. \right\} \text{Spherical Coordinates } [r, \theta, \phi] \quad \begin{cases} 0 \leq r \leq \infty \\ 0 \leq \theta \leq \pi \\ 0 \leq \phi \leq 2\pi \end{cases}$$

→ distance b/w two points: $\vec{p}_2 - \vec{p}_1 = \vec{R} \Rightarrow \text{magnitude: } |\vec{p}_2 - \vec{p}_1|$

→ dot product (projections): $\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \cos\theta$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\text{projection of } \vec{A} \text{ onto } \vec{B}: \quad \underbrace{\vec{A} \cdot \vec{B}}_{\text{magnitude}} \cdot \underbrace{\frac{\vec{B}}{|\vec{B}|}}_{\text{direction}} = \frac{(\vec{A} \cdot \vec{B})}{|\vec{B}|^2} \cdot \vec{B}$$

$$\text{projection of } \vec{B} \text{ onto } \vec{A}: \quad \vec{B}_A = \left[\frac{\vec{B} \cdot \vec{A}}{|\vec{A}|} \right] \cdot \frac{\vec{A}}{|\vec{A}|} = \frac{(\vec{B} \cdot \vec{A})}{|\vec{A}|^2} \cdot \vec{A}$$

→ $\vec{A} \cdot \vec{B} = 0$ ($\vec{A} \perp \vec{B}$)

→ cross product (area): $\vec{A} \times \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \sin\theta \hat{a}_n$

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$$

$$\vec{B} = B_x \hat{x} + B_y \hat{y} + B_z \hat{z}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \hat{x}(A_y B_z - A_z B_y) - \hat{y}(A_x B_z - A_z B_x) + \hat{z}(A_x B_y - A_y B_x)$$

$$\rightarrow \vec{A} \times \vec{B} = 0 \quad \vec{A} \parallel \vec{B}$$

$\rightarrow$ Vector Algebra rules:

$$(1) \vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

$\rightarrow$ Commutative property

$$(2) \vec{A} \times (\vec{B} \times \vec{C}) \neq (\vec{A} \times \vec{B}) \times \vec{C} \quad \rightarrow \text{Associative property}$$

$$(3) \vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$\rightarrow$ distributive property

$$(4) \vec{A} \cdot \vec{A} = |\vec{A}|^2$$

$$\vec{A} \times \vec{A} = 0$$

$\rightarrow$ self product

$\rightarrow$ Scalar Triple product: $(\vec{B} \times \vec{C}) \cdot \vec{A} = (\vec{A} \times \vec{B}) \cdot \vec{C} = (\vec{C} \times \vec{A}) \cdot \vec{B}$
 $\rightarrow$ Volume of parallelepiped

$\rightarrow$ Vector triple product: $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{A} \cdot \vec{C}) - \vec{C} \cdot (\vec{A} \cdot \vec{B})$

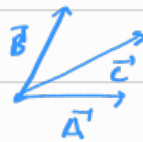
$\rightarrow$ electromagnetic variables are function of time as well as space coordinates

$\rightarrow$ vector can be written as $\vec{A} = a_A |\vec{A}|$

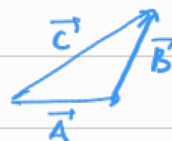
$\rightarrow |\vec{A}| \rightarrow$ magnitude

$a_A = \frac{\vec{A}}{|\vec{A}|} \rightarrow$ direction (dimensionless quantity)

$\rightarrow \vec{A} + \vec{B} = \vec{C} \quad \rightarrow$ (1) parallelogram rule:



$\rightarrow$ (2) head to tail rule:



$\rightarrow \vec{A} + \vec{B} = \vec{B} + \vec{A} \quad \rightarrow$ Commutative law

$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C} \quad \rightarrow$ Associative law

$\rightarrow k \cdot \vec{A} = a_A \cdot (kA)$

$\rightarrow$ SCALAR/DOT PRODUCT: $\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \cos \theta_{AB} \rightarrow$ smaller angle

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} \quad (\text{commutative})$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C} \quad (\text{distributive})$$

$$\rightarrow \text{VECTOR/CROSS PRODUCT: } \vec{A} \times \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \sin \theta_{AB}$$

$$\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$\vec{A} \times (\vec{B} \times \vec{C}) \neq (\vec{A} \times \vec{B}) \times \vec{C}$$

$$\rightarrow \text{SCALAR TRIPLE PRODUCT: } \vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = -\vec{B} \cdot (\vec{A} \times \vec{C}) = -\vec{C} \cdot (\vec{B} \times \vec{A})$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) \Rightarrow \text{Volume of parallelepiped} = |\vec{A} \cdot (\vec{B} \times \vec{C})| = |\vec{B} \times \vec{C}| |\vec{A}| \cos \theta = |\vec{B} \times \vec{C}| |\vec{A}| \sin \theta_1 \cos \theta_2$$

$\underbrace{\quad}_{\text{height}} \quad \underbrace{\quad}_{\text{Area of base}}$

$$\rightarrow \text{VECTOR TRIPLE PRODUCT: } \vec{A} \times (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \cdot \vec{A}) - \vec{C} \cdot (\vec{A} \cdot \vec{B})$$

$$\rightarrow \text{ORTHOGONAL COORDINATE SYSTEM:}$$

$$L \text{ unit vectors } a_{v1}, a_{v2}, a_{v3} \nearrow v_1, v_2, v_3 \rightarrow (x, y, z)$$

$$a_{v1} \times a_{v2} = a_{v3}$$

$$a_{v1} \cdot a_{v2} = a_{v2} \cdot a_{v3} = a_{v3} \cdot a_{v1} = 0$$

$$a_{v2} \times a_{v3} = a_{v1}$$

$$a_{v1} \cdot a_{v1} = a_{v2} \cdot a_{v2} = a_{v3} \cdot a_{v3} = 1$$

$$a_{v3} \times a_{v1} = a_{v2}$$

$$\rightarrow \vec{A} = a_{v1} \cdot |A_{v1}| + a_{v2} \cdot |A_{v2}| + a_{v3} \cdot |A_{v3}|$$

$$|\vec{A}| = \sqrt{A_{v1}^2 + A_{v2}^2 + A_{v3}^2}$$

$$\text{differential length: } dl_i = h_i \cdot du_i \quad \nearrow h_i \rightarrow \text{metric coefficient (f^n of } v_1, v_2, v_3)$$

$$dl = a_{v1} \cdot dl_1 + a_{v2} \cdot dl_2 + a_{v3} \cdot dl_3$$

$$= a_{v1} (h_1 du_1) + a_{v2} (h_2 du_2) + a_{v3} (h_3 du_3)$$

$$|dl| = \sqrt{dl_1^2 + dl_2^2 + dl_3^2}$$

$$dV = dl_1 \cdot dl_2 \cdot dl_3$$

$$= h_1 du_1 \cdot h_2 du_2 \cdot h_3 du_3$$

$$ds_1 = dl_2 \cdot dl_3 = h_2 du_2 \cdot h_3 du_3$$

$$ds_2 = dl_1 \cdot dl_3 = h_1 du_1 \cdot h_3 du_3$$

$$ds_3 = dl_1 \cdot dl_2 = h_1 du_1 \cdot h_2 du_2$$

→ **CARTESIAN COORDINATES:** $\begin{cases} -\infty \leq x \leq \infty \\ -\infty \leq y \leq \infty \\ -\infty \leq z \leq \infty \end{cases}$

$$(u_1, u_2, u_3) = (x, y, z)$$

$$a_x \times a_y = a_z$$

$$a_y \times a_z = a_x$$

$$a_z \times a_x = a_y$$

$$\text{position vector } OP = x_1 a_x + y_1 a_y + z_1 a_z$$

$$\vec{A} = A_x a_x + A_y a_y + A_z a_z$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\vec{A} \times \vec{B} = a_x (A_y B_z - A_z B_y) + a_y (A_z B_x - A_x B_z) + a_z (A_x B_y - A_y B_x)$$

since $h_1, h_2, h_3 = 1$
as x, y, z
are lengths

$$\left[\begin{array}{l} dl = a_x dx + a_y dy + a_z dz \\ ds_x = dy \cdot dz \hat{a}_x \\ ds_y = dx \cdot dz \hat{a}_y \\ ds_z = dx \cdot dy \hat{a}_z \\ dv = dx \cdot dy \cdot dz \end{array} \right.$$

→ **CYLINDRICAL COORDINATES:** $\begin{cases} 0 \leq \rho \leq \infty \\ 0 \leq \phi \leq 2\pi \\ -\infty \leq z \leq \infty \end{cases}$

$$(u_1, u_2, u_3) = (\rho, \phi, z)$$

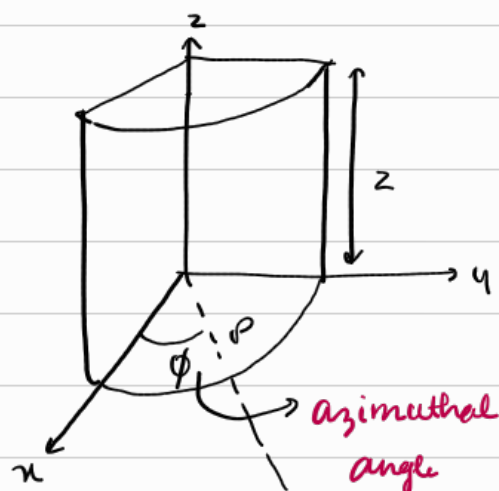
$$a_\rho \times a_\phi = a_z$$

$$a_\phi \times a_z = a_\rho$$

$$a_z \times a_\rho = a_\phi$$

$$A = a_\rho A_\rho + a_\phi A_\phi + a_z A_z$$

ρ and z are lengths ($h_1, h_3 = 1$)



$$dl = a_\rho d\rho + a_\phi \rho d\phi + a_z dz$$

$$ds_\rho = \rho d\phi \cdot dz$$

$$ds_\phi = d\rho \cdot dz$$

$$ds_z = d\rho \cdot \rho d\phi$$

$$dv = \rho d\phi \cdot d\rho \cdot dz$$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

CYLINDRICAL TO CARTESIAN COORDINATES:

$$A_\rho a_\rho + A_\phi a_\phi + A_z a_z \rightarrow A_x a_x + A_y a_y + A_z a_z$$



$$A \cdot a_x = A_\rho \cdot a_\rho \cdot a_x + A_\phi a_\phi \cdot a_x + A_z \cancel{a_z} \cdot a_x$$

$$a_\rho \cdot a_x = \cos \phi$$

$$a_\phi \cdot a_x = \sin(\pi/2 + \phi) = -\sin \phi$$

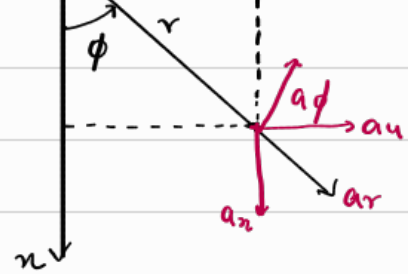
$$A_x = A_\rho \cos \phi - A_\phi \sin \phi$$

$$A \cdot a_y = A_\rho \cdot a_\rho \cdot a_y + A_\phi \cdot a_\phi \cdot a_y + A_z \cdot a_z \cdot a_y$$

$$a_\rho \cdot a_y = \sin(\pi/2 - \phi) = +\sin \phi$$

$$a_\phi \cdot a_y = \cos \phi$$

$$A_y = A_\rho \sin \phi + A_\phi \cos \phi$$



$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

$$x = \rho \cos \theta; y = \rho \sin \theta; z = z$$

$$\rho = \sqrt{x^2 + y^2}; \phi = \tan^{-1}\left(\frac{y}{x}\right); z = z$$

→ SPHERICAL COORDINATES:

$$(v_1, v_2, v_3) = (R, \theta, \phi)$$

$$a_R \times a_\theta = a_\phi$$

$$a_\phi \times a_R = a_\theta$$

$$a_\theta \times a_\phi = a_R$$

$$A = a_R \cdot A_R + a_\phi A_\phi + a_\theta \cdot A_\theta$$

$$h_1 = 1, h_2 = R, h_3 = R \sin \theta$$

$$dl = a_R \cdot dR + a_\theta \cdot R d\theta + a_\phi R \sin \theta d\phi$$

$$ds_R = R d\theta \cdot R \sin \theta d\phi$$

$$ds_\theta = dR \cdot R \sin \theta d\phi$$

$$ds_\phi = dR \cdot R d\theta$$

$$dV = R^2 \sin \theta dR d\phi d\theta$$

SPHERICAL TO CARTESIAN COORDINATES:

$$x = R \sin \theta \cos \phi$$

$$R = \sqrt{x^2 + y^2 + z^2}$$

$$y = R \sin \theta \sin \phi$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$z = R \cos \theta$$

$$y = R \sin \theta \sin \phi$$

$$z = R \cos \theta$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right)$$

$$\phi = \tan^{-1} (y/x)$$

→ INTEGRALS CONTAINING VECTOR FUNCTIONS:

$$\iiint_V \mathbf{F} \cdot d\mathbf{V} \rightarrow \text{volume integral}$$

$$\oint_C \mathbf{V} \cdot d\mathbf{l} \rightarrow \int_{P_1}^{P_2} \mathbf{V} \cdot d\mathbf{l} \rightarrow \oint_C \mathbf{V} \cdot d\mathbf{l} \approx \oint_C \mathbf{V}(x, y, z) \cdot (a_x dx + a_y dy + a_z dz)$$

path of integration

$$\Rightarrow \oint_C \mathbf{V} \cdot d\mathbf{l} \approx a_x \int_C \mathbf{V}(x, y, z) \cdot dx + a_y \int_C \mathbf{V}(x, y, z) \cdot dy + a_z \int_C \mathbf{V}(x, y, z) \cdot dz$$

$$\iint_S \mathbf{A} \cdot d\mathbf{s} \approx \iint_S \mathbf{A} \cdot d\mathbf{s}_n \quad \left\{ d\mathbf{s}_n = \mathbf{a}_n \cdot ds \right.$$

$$\oint_S \mathbf{A} \cdot \mathbf{a}_n \cdot ds \rightarrow \mathbf{a}_n \rightarrow \text{outward from the volume}$$

→ GRADIENT OF A SCALAR FIELD:

$$\nabla_{\text{Cart}} = \frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z$$

$$\nabla_{\text{cyl}} = \frac{\partial}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial}{\partial \phi} \hat{a}_\phi + \frac{\partial}{\partial z} \hat{a}_z$$

$$\nabla_{\text{sph}} = \frac{\partial}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \hat{a}_\phi$$

DIVERGENCE OF VECTOR

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\vec{\nabla} \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

→ DIVERGENCE THEOREM: (GAUSS)

$$\int_S \mathbf{A} \cdot d\mathbf{s} = \int_V (\vec{\nabla} \cdot \vec{A}) \cdot dV$$

→ CURL OF A VECTOR:

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$\rightarrow \nabla^2 V \rightarrow \text{Laplacian}$$

$$\rightarrow \vec{\nabla} \cdot \vec{A} = 0 \quad (\text{solenoidal})$$

$$\vec{\nabla} \times \vec{A} = 0 \quad (\text{irrotational})$$

$$\vec{\nabla} \times \vec{A} = \frac{1}{\rho} \begin{vmatrix} a_\rho & \rho a_\phi & a_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\vec{\nabla} \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} a_r & r a_\theta & r \sin \theta a_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

→ LINE INTEGRAL:



$$\int_L \vec{A} \cdot d\vec{l} = \int_a^b |\vec{A}| \cos \theta \cdot dl \rightarrow \text{line integral}$$

$$\Psi = \int_S |\vec{A}| \cos \theta \cdot ds = \int_S \vec{A} \cdot \hat{n} \cdot ds \rightarrow \text{surface integral}$$

$$\int_V \rho_v dv \rightarrow \text{volume integral}$$

↖ scalar

→ closed path defines an open surface whereas closed surface defines a volume

→ STOKES THEOREM:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} \cdot ds$$

COULUMB'S LAW: ↗ experimental law

→ $1C \Rightarrow 6 \times 10^{18} e^-$

→ $1e^- = 1.6019 \times 10^{-19} C$

→ force b/w two point charges Q_1 and Q_2

(a) along the line joining them

(b) directly proportional to the product of $Q_1 Q_2$

(c) inv proportional to the square of the distance R b/w them

$$F \propto \frac{Q_1 Q_2}{R^2} \Rightarrow F = k \frac{Q_1 Q_2}{R^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$k \approx 9 \times 10^9 \text{ N/F}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{R^2}$$

$$\begin{aligned} \text{on } Q_2 \text{ due to } Q_1 \quad F_{12} &= \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} a_{R_{12}} \\ &\Rightarrow \frac{Q_1 Q_2 \vec{R}_{12}}{4\pi\epsilon_0 R^3} \end{aligned}$$

$$R_{12} = \vec{r}_2 - \vec{r}_1$$

$$R = |R_{12}|$$

$$a_{R_{12}} = \frac{R_{12}}{|R_{12}|} = \frac{R_{12}}{R}$$

$$F_{12} = \frac{Q_1 Q_2 (\vec{r}_2 - \vec{r}_1)}{4\pi\epsilon_0 |\vec{r}_2 - \vec{r}_1|^3}$$

$$F_{21} = \frac{Q_1 Q_2 (\vec{r}_1 - \vec{r}_2)}{4\pi\epsilon_0 (\vec{r}_1 - \vec{r}_2)^3} = -F_{12}$$

→ like charges repel while unlike charges attract

→ distance R b/w Q_1 and Q_2 must be comparable to the dimensions of Q_1 and Q_2

→ Q_1 and Q_2 must be static (at rest)

→ quantity of total charge remains constant

ELECTRIC FIELD INTENSITY: $\nearrow$ is that force that a unit positive charge experiences when placed in an electric field

$$E = \lim_{Q \rightarrow 0} \frac{F}{Q} \Rightarrow E = \frac{F}{Q} \text{ N/C (or) V/m}$$

$$E = \frac{Q (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} = \frac{Q}{4\pi\epsilon_0 r^2} a_r$$

$$E = E_1 + E_2 + E_3 + \dots + E_n$$

$$E = \frac{Q_1 (\vec{r} - \vec{r}_1)}{4\pi\epsilon_0 (\vec{r} - \vec{r}_1)^3} + \frac{Q_2 (\vec{r} - \vec{r}_2)}{4\pi\epsilon_0 (\vec{r} - \vec{r}_2)^3} + \dots + \frac{Q_n (\vec{r} - \vec{r}_n)}{4\pi\epsilon_0 (\vec{r} - \vec{r}_n)^3}$$

→ **ELECTRIC FIELDS DUE TO CONTINUOUS CHARGE DISTRIBUTIONS:**

→ a line, surface (or) volume will have continuous charge distribution

→ line charge density $\Rightarrow \rho_L$ (C/m)

→ surface charge density $\Rightarrow \rho_s$ (C/m²)

→ volume charge density ρ_v (C/m^3)

$$dQ = \rho_l \cdot dl \quad [C/m \times m] \rightarrow \text{small charge element}$$

$$\text{total charge: } Q = \int_L \rho_l \cdot dl$$

$$dQ = \rho_s \cdot ds \Rightarrow Q = \int_S \rho_s \cdot ds$$

$$dQ = \rho_v \cdot dv \Rightarrow Q = \int_V \rho_v \cdot dv$$

$$\text{line charge: } E = \frac{1}{4\pi\epsilon_0} \int_L \frac{\rho_l \cdot dl \cdot a_R}{R^2}$$

$$\text{surface charge: } E = \frac{1}{4\pi\epsilon_0} \int_S \frac{\rho_s \cdot ds \cdot a_R}{R^2}$$

$$\text{volume charge: } E = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_v \cdot dv \cdot a_R}{R^2}$$

→ R^2 and a_R vary as the integrals are evaluated

→ ELECTRIC FLUX DENSITY:

↳ value of E depends on the medium (air, vacuum, glass etc)

→ E changes as the material changes

→ new field $\Rightarrow D = \epsilon_0 E$

→ $D \rightarrow$ amt of electric flux/unit area [C/m^2]

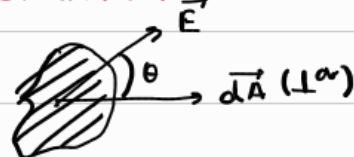
$$\rightarrow \Psi = \int_S D \cdot ds$$

→ one line of Ψ emanates from $+1C$ and terminate on $-1C$

$$(eq) \quad D = \epsilon_0 E$$

$$= \epsilon_0 \int \frac{\rho_s \cdot ds \cdot a_R}{4\pi\epsilon_0 R^2} \checkmark \rightarrow \text{using gauss law} \Rightarrow \frac{\rho_s}{2}$$

ELECTRIC FLUX:



$$d\phi = \vec{E} \cdot d\vec{A} = E dA \cdot \cos\theta$$

$$\phi = \int E \cdot d\vec{A} \cdot \cos\theta$$

$$\phi = \vec{E} \cdot \vec{A}$$

$$\phi = \int E \cdot d\vec{s}$$

→ GAUSS LAW - MAXWELL EQUATION:

Gauss law states that the total electric flux through any closed surface is equal to the total charge enclosed by that surface $\times 1/\epsilon_0$

$$\Psi = Q_{enc} / \epsilon_0$$

$$\Psi = \oint_S \vec{D} \cdot d\vec{A} = \oint_S \rho_v \cdot dV$$

$$Q = \int_V \rho_v \cdot dV$$

$$Q = \Psi = \int_V \rho_v \cdot dV = \oint_S \vec{D} \cdot d\vec{A}$$

$$\Rightarrow \oint_S \vec{D} \cdot d\vec{A} = \int_V \vec{\nabla} \cdot \vec{D} \cdot dV = \int_V \rho_v \cdot dV$$

$$\therefore \rho_v = \vec{\nabla} \cdot \vec{D}$$

$$\Phi = \int \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$$\frac{Q_{enc}}{\epsilon_0} = \int \vec{E} \cdot d\vec{A} = Q_{enc} = \oint \vec{D} \cdot d\vec{A}$$

→ size independent → flux passes through all the closed surface of the sphere

→ shape independent → does not depend on the dimension



$$Q_{net} = q_1 + q_2 + q_3$$

$$\Phi_{net} = \oint \vec{E} \cdot d\vec{A} \rightarrow E \text{ calculated w.r.t to } q_1, q_2 \text{ and } q_3$$

→ APPLICATIONS OF GAUSS LAW:

↳ check if symmetry exists

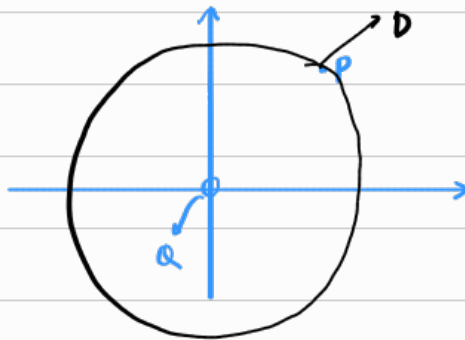
↳ construct a mathematical closed surface (Gaussian surface)

↳ choose the surface such that $\vec{D}$ is normal/tangential to the surface

$$\vec{D} \cdot d\vec{A} = |\vec{D}| \cdot |d\vec{A}| \rightarrow \text{when } \vec{D} \text{ is } \perp^{\text{ar}}$$

$$\vec{D} \cdot d\vec{A} = 0 \rightarrow \text{when } \vec{D} \text{ is tangential}$$

(A) Point charge:



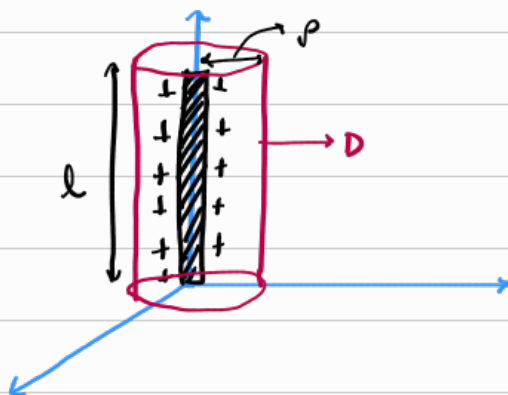
$$\vec{D} = D_r \cdot \hat{a}_r \quad (\vec{D} \text{ is normal to the surface})$$

$$Q = \oint \vec{D} \cdot d\vec{A} = D_r \oint dA = D_r \cdot 4\pi r^2$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0} \Rightarrow E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

(B) Infinite line charge:



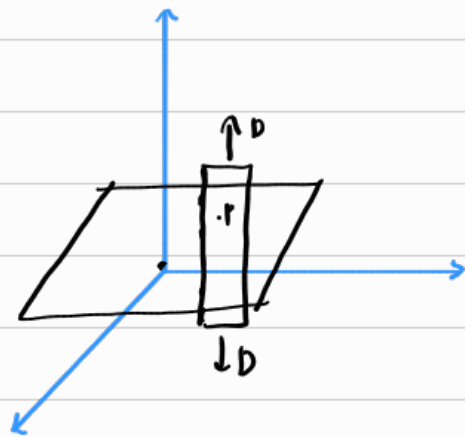
$$\rho_l \cdot dl = Q = \oint \vec{D} \cdot d\vec{A} = D_\rho \oint dA = D_\rho \cdot 2\pi\rho l$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0} \Rightarrow E \cdot 2\pi\rho l = \frac{Q}{\epsilon_0}$$

$$E = \frac{\rho_l \cdot l}{2\pi\epsilon_0 \rho l} = \boxed{\frac{E \propto 1}{\rho}}$$

$$Q = \int \rho_l \cdot dl = \rho_l \cdot L$$

(c) Infinite sheet of charge:



$$\rho_s \cdot ds \Rightarrow \oint \vec{D} \cdot d\vec{s} = D_z \oint_{\text{top}} ds + D_z \oint_{\text{bottom}} ds$$

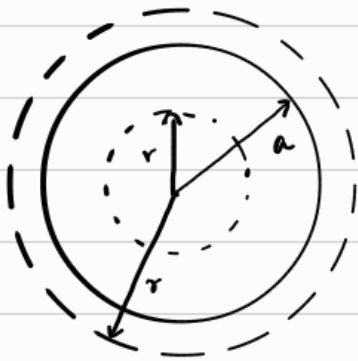
$$\approx D_z (A + A) = D_z (2A) = \rho_s \cdot A$$

$$\int \vec{E} \cdot d\vec{A} \approx \frac{Q}{\epsilon_0} \approx E \cdot 2A = \frac{Q}{\epsilon_0}$$

$$E \approx \frac{Q}{2A\epsilon_0} = \frac{\rho_s A}{2A\epsilon_0} = \frac{\rho_s}{2\epsilon_0}$$

$$Q = \int \rho_s \cdot ds = \rho_s \cdot A$$

(d) Uniformly Charged sphere:



CASE 1: \$r \le a\$

$$\int \vec{E} \cdot d\vec{A} \approx \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$Q_{\text{enc}} = \int \rho_v \cdot dV \approx \rho_v \int_0^r \int_0^{2\pi} \int_0^\pi r^2 \sin\theta \cdot d\theta d\phi dr$$

$$\approx \rho_v \cdot \frac{4\pi r^3}{3} \approx D_r 4\pi r^2$$

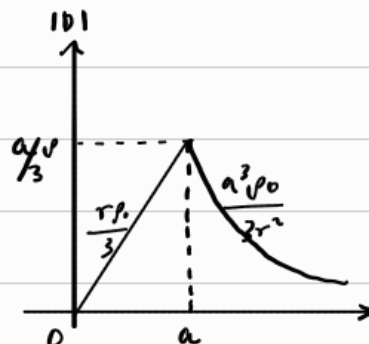
$$D_r \approx \frac{\rho_v r \cdot 4\pi r}{3} \quad 0 \leq r \leq a$$

CASE 2: \$r \ge a\$

$$\int \vec{E} \cdot d\vec{A} \approx \frac{Q_{\text{enc}}}{\epsilon_0} \Rightarrow E \cdot 4\pi r^2 = \frac{4\pi a^3}{3 \cdot \epsilon_0} \approx E \approx \frac{a^3}{3r^2 \epsilon_0}$$

$$D_r 4\pi r^2 = \frac{4\pi a^3}{3} \rho_v \approx D_r \approx \frac{a^3}{3r^2} \rho_v \cdot 4\pi$$

$$D = \begin{cases} \frac{r}{3} \rho_v 4\pi & 0 < r \leq a \\ \frac{a^3}{3r^2} \rho_v 4\pi & r \geq a \end{cases}$$



(E) Electric field due to cylinders:

Solid conducting/Hollow conducting
[charge resides on the surface]

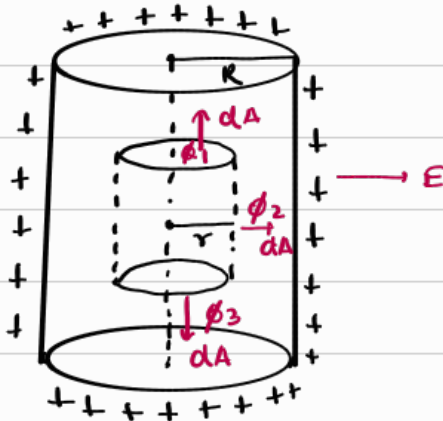
$$\rho_s = \frac{C}{m^2}$$

Solid non conducting/Insulator
[charge can be inside also]

$$\rho_v = \frac{C}{m^3}$$

(1) EF due to solid conducting/HC/HNC cylinder

CASE I: $r < R$



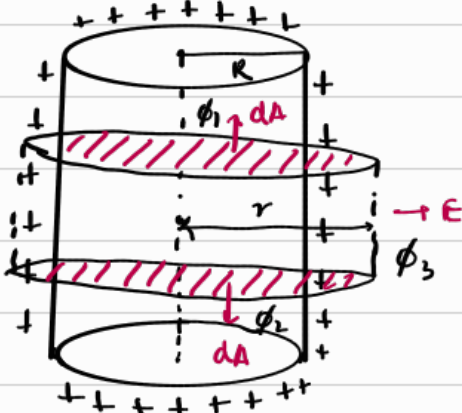
$$\phi_T = \phi_1 + \phi_2 + \phi_3$$

$$\phi_1 = \phi_3 = 0 \quad [\vec{E} \cdot d\vec{A} \cos 90^\circ = 0]$$

$$\phi_2 \Rightarrow \int \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0} \quad \text{as } Q_{enc} = 0 \text{ as charge lies on the surface}$$

$$\therefore E = 0$$

CASE II: $r > R$



$$\phi_T = \phi_2$$

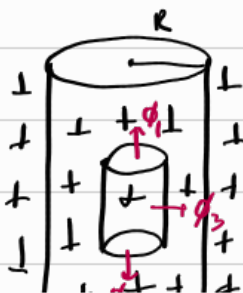
$$\Rightarrow \int \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$$\Rightarrow E \cdot 2\pi r L = \frac{Q_{enc}}{\epsilon_0} \Rightarrow E = \frac{Q_{enc}}{2\pi r L \epsilon_0}$$

$$\rho_s \Rightarrow \frac{Q_{enc}}{2\pi R L} \Rightarrow Q_{enc} = \rho_s \cdot 2\pi R L$$

$$\therefore \vec{E} = \frac{\rho_s \cdot 2\pi R L}{2\pi r L \epsilon_0} = \frac{\rho_s R}{r \epsilon_0} \hat{a}_r \Rightarrow \vec{E} \propto \frac{1}{r}$$

(2) Solid non conducting cylinder



CASE I: $r < R$

$$\phi_T = \phi_1 + \phi_2 + \phi_3$$

$$\frac{Q_{inside}}{\epsilon_0} = E \cdot 2\pi r L$$



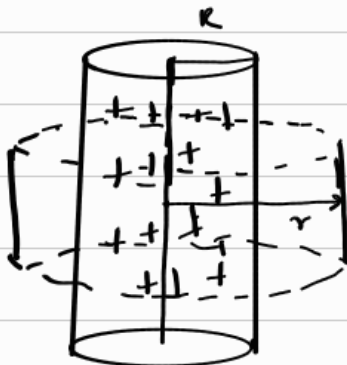
$$E \Rightarrow \frac{Q_{\text{inside}}}{\epsilon_0 \cdot 2\pi r L}$$

$$\rho_V \Rightarrow \frac{Q_{\text{inside}}}{\pi r^2 L}$$

$$\rho_V \times \pi r^2 L = Q_{\text{inside}}$$

$$\vec{E} \Rightarrow \frac{\rho_V \cdot \pi r^2 L}{\epsilon_0 \cdot 2\pi r L} = \frac{\rho_V r}{2\epsilon_0} \hat{a}_r \Rightarrow \boxed{E \propto r}$$

CASE 2: $r > R$



$$\phi_1 = \phi_2$$

$$\frac{Q_{\text{inside}}}{\epsilon_0} = E \times 2\pi r \cdot L$$

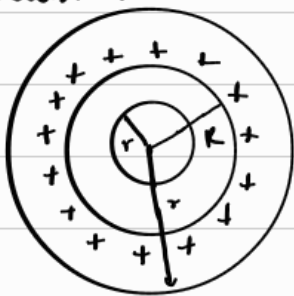
$$\rho_V = \frac{Q_{\text{inside}}}{\pi R^2 L}$$

$$\rho_V \cdot \pi R^2 L = Q_{\text{inside}}$$

$$\therefore E \Rightarrow \frac{\rho_V \cdot \pi R^2 L}{\epsilon_0 \cdot 2\pi r L} = \frac{\rho_V R^2}{2\epsilon_0 r} \hat{a}_r \Rightarrow \boxed{E \propto \frac{1}{r}}$$

(F) Solid Conducting/Hollow Conducting/Hollow non conducting sphere:

(i) $r > R$ (Outside)



$$\phi \Rightarrow \int \vec{E} \cdot d\vec{A}$$

$$= E \cdot 4\pi r^2 \Rightarrow \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$Q_{\text{enc}} \Rightarrow \rho_s \cdot 4\pi R^2$$

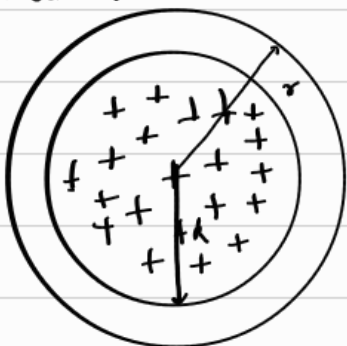
$$E \Rightarrow \frac{\rho_s \cdot 4\pi R^2}{4\pi r^2 \cdot \epsilon_0} = \frac{\rho_s \cdot R^2}{r^2 \cdot \epsilon_0} \hat{a}_r \Rightarrow \boxed{E \propto \frac{1}{r^2}}$$

(ii) $r < R$ (Inside)

$$\phi = \int \vec{E} \cdot d\vec{A} = E \cdot 4\pi r^2 = \frac{Q_{\text{enc}}}{\epsilon_0} = 0$$

solid non conducting

(i) $r > R$ (Outside)



$$\int \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0} \Rightarrow E \cdot 4\pi r^2 = \frac{\rho_V \cdot 4\pi R^3}{3\epsilon_0}$$

$$E \Rightarrow \frac{\rho_V \cdot R^3}{3r^2 \cdot \epsilon_0} \Rightarrow \boxed{E \propto \frac{1}{r^2}}$$

(ii) $r < R$ (inside)

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0} \Rightarrow E \cdot 4\pi r^2 = \frac{\rho_v \cdot \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$E = \frac{\rho_v \cdot r}{3\epsilon_0} \quad \text{or } \boxed{E dr}$$

→ ELECTRIC POTENTIAL:

→ moving a charge Q from A to B in an electric field E

$$\rightarrow F = QE$$

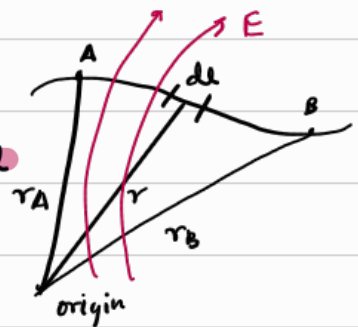
$$\rightarrow dW \approx \ominus F \cdot dl \Rightarrow -QE \cdot dl$$

work done
by ext agent

$$\rightarrow W \approx -Q \int E \cdot dl$$

$$\left\{ \frac{W}{Q} = - \int E \cdot dl \Rightarrow V_{AB} = - \int E \cdot dl \right.$$

potential energy per unit charge



→ A → initial point B → final point

→ $V_{AB} \rightarrow -ve$ (work is done by the field)

$V_{AB} \rightarrow +ve$ (ext agent does work)

→ $V_{AB} \rightarrow$ independent of the path taken

→ $V_{AB} \rightarrow$ measured in J/C (or) Volts (V)

$$E \approx \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \Rightarrow V_{AB} = - \int_{r_A}^{r_B} \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \cdot dr \hat{r}$$

$$\int \frac{dr}{r^2} = -\frac{1}{r}$$

$$\approx \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r_B} - \frac{1}{r_A} \right]$$

$$V_{BA} \approx \frac{Q}{4\pi\epsilon_0 r_B} - \frac{Q}{4\pi\epsilon_0 r_A} \approx V_{AB} = V_B - V_A$$

absolute potentials of B and A

→ potential at infinity assumed to be 0 [$r_A \rightarrow \infty$]

$$V \approx \frac{Q}{4\pi\epsilon_0 r}$$

$$\therefore V \approx - \int E \cdot dl \quad \text{suppose point charge not located at origin}$$

∞

$$\rightarrow V(r) = \frac{Q}{4\pi\epsilon_0 |r-r'|}$$

superposition with many point charges:

$$\rightarrow V(r) = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^n \frac{Q_k}{|r-r_k|} \rightarrow \text{for discrete point charges}$$

$$\rightarrow V(r) = \frac{1}{4\pi\epsilon_0} \int_L \frac{\rho_L(r') \cdot dl'}{|r-r'|} \rightarrow \text{for continuous line charge}$$

$$V(r) = \frac{1}{4\pi\epsilon_0} \int_S \frac{\rho_S(r') \cdot ds'}{|r-r'|} \rightarrow \text{for continuous surface charge}$$

$$V(r) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_V(r') \cdot dv'}{|r-r'|} \rightarrow \text{for continuous volume charge}$$

$\rightarrow$ If the 0 potential is not chosen as infinity then

$$V = \frac{Q}{4\pi\epsilon_0 r} + c \quad \text{chosen point of ref}$$

$$V = - \int E \cdot dl + c$$

$$V_{AB} = V_B - V_A = - \int_A^B E \cdot dl = \frac{W}{Q}$$

$\rightarrow$ RELATIONSHIP B/W E AND V $\rightarrow$ MAXWELL'S EQUATION:

$$V_{AB} = -V_{BA}$$

$$V_{AB} + V_{BA} = \oint_C E \cdot dl = 0$$

$\oint_C E \cdot dl = 0 \rightarrow$ line integral of E along a closed path must be 0

$\hookrightarrow$ no net work is done in moving a charge along a closed path in an electrostatic field

Applying Stokes theorem:

$$\oint_C E \cdot dl = \int_S (\nabla \times E) \cdot ds = 0$$

$\therefore \nabla \times E = 0 \rightarrow$ conservative/irrotational

$$\left. \begin{array}{l} \oint_C E \cdot dl = 0 \\ \nabla \times E = 0 \end{array} \right\} \rightarrow \text{Maxwell's equation for static electric fields}$$

$$\rightarrow V = - \int E \cdot dl$$

$$dV = -E \cdot dl = -E_x dx - E_y dy - E_z dz$$

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz$$

$$\frac{\partial V}{\partial x} + \frac{\partial V}{\partial y} + \frac{\partial V}{\partial z}$$

$$E_x = -\frac{\partial V}{\partial x} ; E_y = -\frac{\partial V}{\partial y} ; E_z = -\frac{\partial V}{\partial z}$$

$$\mathbf{E} = -\nabla V$$

→ ELECTRIC POTENTIAL DUE TO POINT CHARGE Q

$$Q \xrightarrow{r} P \quad \left[\vec{E} = \frac{1 \cdot Q}{4\pi\epsilon_0 \cdot r^2} \hat{a}_r \right]$$

$$V_P = \vec{E} \cdot \vec{r} = \frac{Q}{4\pi\epsilon_0 r^2} \cdot r = \frac{Q}{4\pi\epsilon_0 r}$$

$$V_{AB}(\text{charge}) = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_B} - \frac{1}{r_A} \right)$$

$$V_{AB}(\text{line}) \Rightarrow \vec{E} = \int \frac{\rho_L \cdot d\vec{L}}{4\pi\epsilon_0 r^2} \Rightarrow \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$$V_{AB} = - \int_{r_B}^{r_A} \frac{\rho_L}{2\pi\epsilon_0 r} \cdot d\vec{L} = - \frac{\rho_L}{2\pi\epsilon_0} \ln r \Big|_{r_B}^{r_A}$$

$$\vec{E} \cdot 2\pi r L = \frac{\rho_L \cdot L}{\epsilon_0} \Rightarrow \vec{E} = \frac{\rho_L}{2\pi\epsilon_0 r} \hat{a}_r$$

$$= - \frac{\rho_L}{2\pi\epsilon_0} \ln \left(\frac{r_A}{r_B} \right)$$

$$V_{AB}(\text{sheet}) \Rightarrow \vec{E} = \int \frac{\rho_s \cdot d\vec{s}}{4\pi\epsilon_0 r^2} \Rightarrow \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0} = \vec{E} \cdot 2A = \frac{\rho_s \cdot A}{\epsilon_0} \Rightarrow \vec{E} = \frac{\rho_s}{2\epsilon_0} \hat{a}_r$$

$$V_{AB} = - \int_{x_B}^{x_A} \frac{\rho_s}{2\epsilon_0} \cdot d\vec{L} = - \frac{\rho_s}{2\epsilon_0} [x_A - x_B]$$

→ POISSON'S AND LAPLACE EQUATIONS:

$$\vec{\nabla} \cdot \vec{D} = \rho_V \Rightarrow \vec{\nabla} \cdot (\epsilon_0 \vec{E})$$

$$\mathbf{E} = -\nabla V$$

$$-\vec{\nabla} \cdot (\epsilon_0 \cdot \nabla V) = \rho_V \rightarrow \text{considering } \epsilon_0 \rightarrow \text{constant you can divide it}$$

$$\nabla^2 V = -\frac{\rho_V}{\epsilon_0} \rightarrow \text{Poisson's equation}$$

$$\rightarrow \text{when } \rho_V = 0 \Rightarrow \nabla^2 V = 0 \rightarrow \text{Laplace's equation}$$

$$\text{LAPLACE EQUATION: Cartesian coordinates: } \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

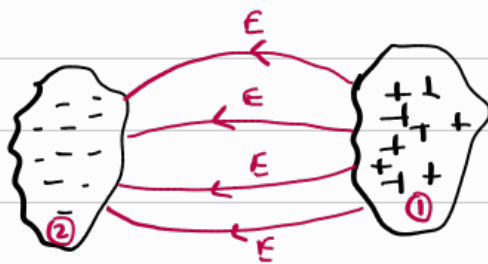
Cylindrical coordinates: $\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$

Spherical coordinates: $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$

$$\frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

→ CAPACITANCE:

- ↳ two/more conductors carrying equal but opposite charges
- ↳ flux leaving one conductor must terminate at the surface of the other conductor
- ↳ conductors $\Rightarrow$ plates of the capacitor



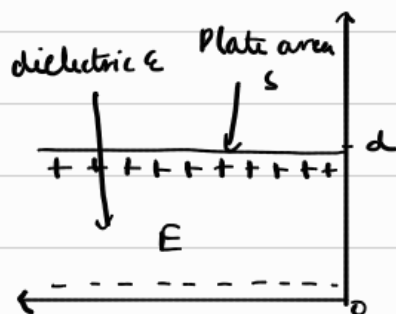
$$V = V_1 - V_2 = - \int_1^2 \mathbf{E} \cdot d\mathbf{l}$$

$$C = \frac{Q}{V} \quad \begin{array}{l} \text{(carried by one plate)} \\ \text{(potential diff b/w them)} \end{array}$$

$$C = \frac{Q}{V} = \frac{\epsilon \oint \mathbf{E} \cdot d\mathbf{s}}{\int \mathbf{E} \cdot d\mathbf{l}}$$

→ $C \rightarrow$ physical property of the capacitor, measured in Farads (F)

(A) Parallel-Plate Capacitor:



$\rho_s = \frac{Q}{s}$ → ideal parallel plate capacitor is where d is very small compared to the dimensions of the plate

$$C = \frac{Q}{V} \quad \begin{array}{l} \text{At } x=0 \quad V=0 \\ \text{At } x=d \quad V=V_0 \end{array}$$

For dielectric $\rho_v = 0$

$$\nabla^2 V = 0$$

$$\frac{\partial^2 V}{\partial x^2} = 0 \Rightarrow \frac{\partial V}{\partial x} = \omega x + t \quad \Rightarrow V = \Delta x + B$$

$$x=0 \quad [0 = B]$$

$$x=d \quad V_0 = Ad + 0 \Rightarrow \boxed{A = \frac{V_0}{d}}$$

$$V \propto \frac{V_0 x}{d}$$

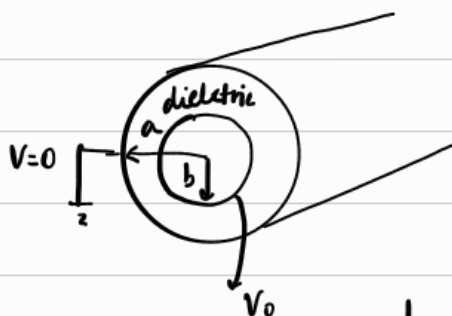
$$E \propto -\nabla V \propto -\frac{\partial V}{\partial x} \hat{a}_x \propto -\frac{V_0}{d} \hat{a}_x \text{ V/m}$$

$$\vec{D} = \epsilon_0 \vec{E} = -\frac{\epsilon_0 V_0}{d} \hat{a}_x \text{ C/m}^2$$

$$C = \frac{Q}{V} \Rightarrow \left| \frac{D \cdot A}{V_0} \right| = \left| \frac{-\epsilon_0 V_0 A}{d V_0} \right| = \left| \frac{-\epsilon_0 A}{d} \right| = \frac{\epsilon_0 A}{d}$$

$$W_E = \frac{1}{2} CV^2 = \frac{1}{2} QV^2 = \frac{Q^2}{2C}$$

(B) Cylindrical Capacitor / Coaxial Cable:



→ inner capacitor taken as infinite line of charge
so only radial field exists

$\nabla \rightarrow f''$ of r only

$\therefore \nabla^2 V = 0$ (Laplace equation)

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

$$\Rightarrow \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = 0$$

$$\int \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = \int 0 \quad \therefore r \frac{\partial V}{\partial r} = A$$

$$\int \partial V = \int \frac{A \partial r}{r}$$

$$\boxed{V \propto A \ln r + B}$$

$$\text{at } r=a; V=0$$

$$r=b; V=V_0$$

$$\text{at } r=a \Rightarrow 0 = A \ln a + B$$

$$\boxed{B = -A \ln a}$$

$$\text{at } r=b \Rightarrow V_0 = A \ln b - A \ln a$$

$$V_0 = A \ln \left(\frac{b}{a} \right)$$

$$\boxed{A = V_0}$$

$$\Rightarrow V = A \ln r + B$$

$$\ln(b/a)$$

$$V = \frac{V_0 \ln r}{\ln(b/a)} - \frac{V_0 \ln(a)}{\ln(b/a)}$$

$$V = \frac{V_0}{\ln(b/a)} \cdot \ln(r/a)$$

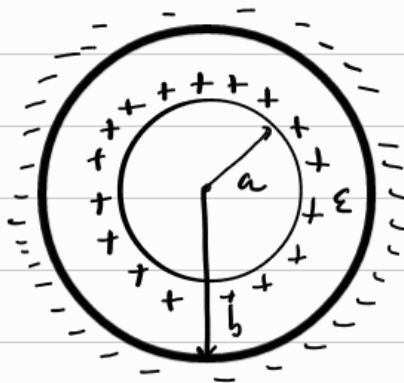
$$E = -\frac{\partial V}{\partial r} = -\frac{1}{r} \frac{V_0}{\ln(b/a)} = -\frac{A}{r}$$

$$Q = \iint \mathbf{D} \cdot d\mathbf{s} = \iint \epsilon_0 \frac{V_0}{\ln(b/a)} \cdot \frac{1}{r} \cdot r d\phi dz$$

$$= \frac{\epsilon_0 V_0}{\ln(b/a)} \iint d\phi dz = \frac{\epsilon_0 V_0}{\ln(b/a)} 2\pi h = \frac{V_0 2\pi h \epsilon_0}{\ln(b/a)}$$

$$C = \frac{Q}{V} = \frac{V_0 2\pi h \epsilon_0}{V_0 \ln(b/a)} = \frac{\epsilon_0 2\pi h}{\ln(b/a)} = C$$

(C) Spherical Capacitor:



$$a < r < b$$

$$Q = \epsilon \int \mathbf{E} \cdot d\mathbf{s} = \epsilon \cdot E_r \cdot 4\pi r^2$$

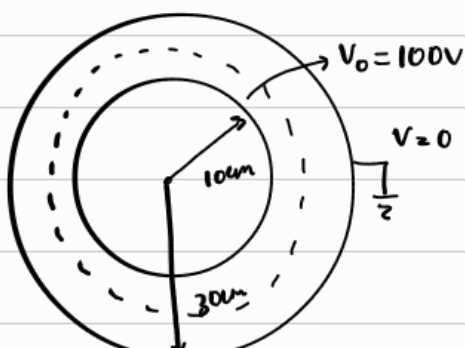
$$E = \frac{Q}{4\pi \epsilon_0 r^2}$$

$$V = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \int_b^a \frac{Q}{4\pi \epsilon_0 r^2} \cdot dr$$

$$= \frac{Q}{4\pi \epsilon_0} \left[\frac{1}{a} - \frac{1}{b} \right]$$

$$C = \frac{Q}{V} = \frac{4\pi \epsilon_0}{\left[\frac{1}{a} - \frac{1}{b} \right]}$$

(eg)



$$\epsilon_r = 2.5$$

$$10 < r < 30$$

$$\vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0} \Rightarrow Q_{enc} = \epsilon \cdot \vec{D} \cdot d\vec{A}$$

$$= \epsilon \cdot D_r \cdot 4\pi r^2$$

$$\vec{E} = \frac{Q_{enc}}{4\pi r^2 \epsilon}$$

$$\nabla^2 V = 0 \Rightarrow \frac{1}{r^2} \left(\frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) \right) = 0$$

$$\Rightarrow r^2 \frac{\partial V}{\partial r} = A \Rightarrow \frac{\partial V}{\partial r} = \frac{A}{r^2}$$

$$\Rightarrow V = -\frac{A}{r} + B$$

when $r = b$ $V = 0$

$$0 = -\frac{A}{b} + B \Rightarrow \boxed{B = \frac{A}{b}}$$

when $r = a$ $V = V_0$

$$V_0 = -\frac{A}{a} + \frac{A}{b} \Rightarrow V_0 = A \left[\frac{1}{b} - \frac{1}{a} \right]$$

$$A = \frac{V_0}{\left[\frac{1}{b} - \frac{1}{a} \right]}$$

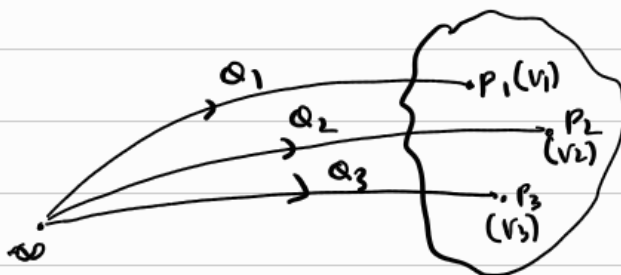
$$\Rightarrow V = \frac{V_0}{\left(\frac{1}{a} - \frac{1}{b} \right) r} + \frac{V_0}{\left(\frac{1}{b} - \frac{1}{a} \right) b} = \frac{V_0 \left[\frac{1}{b} - \frac{1}{r} \right]}{\left[\frac{1}{b} - \frac{1}{a} \right]}$$

$$E = -\nabla V = -\frac{\partial V}{\partial r} = \frac{V_0}{\left(\frac{1}{b} - \frac{1}{a} \right) r^2}$$

$$Q = \vec{E} \cdot d\vec{s} \cdot \epsilon = \frac{+V_0}{\left(\frac{1}{a} - \frac{1}{b} \right) r^2} \cdot 4\pi r^2 \cdot 2.5$$

$$C = \frac{Q}{V} = \frac{\frac{V_0}{\left(\frac{1}{a} - \frac{1}{b} \right) \cdot V_0} \cdot 4\pi \times 25}{\left(\frac{1}{10} - \frac{1}{30} \right)} =$$

→ ENERGY DENSITY IN ELECTROSTATIC FIELDS:



$$W_E \approx W_1 + W_2 + W_3$$

$$\approx 0 + Q_2(V_{21}) + Q_3(V_{31} + V_{32})$$

$$W_E \approx W_3 + W_2 + W_1$$

$$\approx 0 + Q_2(V_{23}) + Q_1(V_{13} + V_{12})$$

$$2W_E = Q_1V_1 + Q_2V_2 + Q_3V_3$$

$$W_E = \frac{1}{2} [Q_1V_1 + Q_2V_2 + Q_3V_3] = \frac{1}{2} \sum_{k=1}^n Q_k \cdot V_k \rightarrow \text{for discrete point charges}$$

→ for continuous charge distribution:

$$W_E = \frac{1}{2} \int \rho_L \cdot dL \cdot V \rightarrow \text{line charge}$$

$$W_E = \frac{1}{2} \int \rho_S \cdot dS \cdot V \rightarrow \text{surface charge}$$

$$W_E = \frac{1}{2} \int \rho_V \cdot dV \cdot V \rightarrow \text{volume charge}$$

Since $\rho_V = \nabla \cdot \vec{D}$

$$W_E = \frac{1}{2} \int (\nabla \cdot \vec{D}) V \cdot dV$$

$V \rightarrow$ scalar; $\nabla \cdot \vec{D} \rightarrow$ vector for some scalar V , vector $\vec{A}$

$$\nabla \cdot (V \cdot \vec{A}) = \vec{A} \cdot \nabla V + V \nabla \cdot \vec{A} \quad (\text{or})$$

$$(\nabla \cdot \vec{A}) \cdot V = \nabla \cdot (V \vec{A}) - \vec{A} \cdot \nabla V$$

$$W_E = \frac{1}{2} \int \nabla \cdot (V \vec{D}) \cdot dV = \frac{1}{2} \int \vec{D} \cdot (\nabla V) \cdot dV$$

Apply divergence theorem to the 1st term of R.H.S.

$$W_E = \frac{1}{2} \oint_S (V \vec{D}) \cdot d\vec{s} - \frac{1}{2} \int (\vec{D} \cdot \nabla V) \cdot dV$$

$V \propto 1/r^2 \rightarrow$ varies atleast as $1/r^3$
 $d\vec{s} \rightarrow r^2 \rightarrow$ tends to 0 when surface S becomes large

$$W_E = -\frac{1}{2} \int \vec{D} \cdot \nabla V \cdot dV = \frac{1}{2} \int \vec{D} \cdot \vec{E} \cdot dV$$

$$\vec{D} = \epsilon_0 \epsilon_r \cdot \vec{E}$$

$$\vec{E} = -\nabla V$$

$$= \frac{1}{2} \int_V \epsilon_0 \cdot \epsilon_r E^2 \cdot dV$$

Energy density: $w_e = \frac{dW_e}{dV} = \frac{1}{2} \frac{\rho \cdot E}{\epsilon_0} = \frac{1}{2} \epsilon_0 \epsilon_r E^2 = \frac{\rho^2}{2\epsilon_0} \quad (\text{J/m}^3)$

→ BOUNDARY CONDITIONS:

↳ if the field exists in a region consisting of two diff media, the conditions that the field must satisfy at the interface separating the media are called boundary conditions

→ helpful for determining the field on one side of the boundary if the field on the other side is known

→ to determine boundary conditions, we need to use Maxwell's equations

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$$

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q_{enc}$$

$\mathbf{E} \Rightarrow \mathbf{E}_t + \mathbf{E}_n \rightarrow \mathbf{E}_t \rightarrow$ tangential component of $\mathbf{E}$

$\mathbf{E}_n \rightarrow$ normal component of $\mathbf{E}$

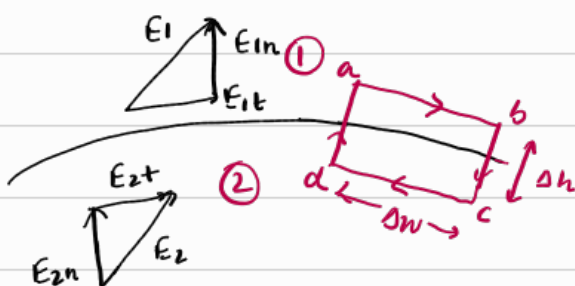
$\mathbf{D} \Rightarrow \mathbf{D}_t + \mathbf{D}_n \rightarrow \mathbf{D}_t \rightarrow$ tangential component of $\mathbf{D}$

$\mathbf{D}_n \rightarrow$ normal component of $\mathbf{D}$

(A) Dielectric - dielectric boundary conditions:

↳ $\mathbf{E}_1 \Rightarrow \mathbf{E}_{1t} + \mathbf{E}_{1n} \rightarrow$ dielectric 1 $\epsilon_1 \Rightarrow \epsilon_0 \epsilon_{r1}$

$\mathbf{E}_2 \Rightarrow \mathbf{E}_{2t} + \mathbf{E}_{2n} \rightarrow$ dielectric 2 $\epsilon_2 \Rightarrow \epsilon_0 \epsilon_{r2}$



$$\begin{aligned} \oint_C \mathbf{E} \cdot d\mathbf{l} &= 0 \\ \Rightarrow 0 &\Rightarrow E_{1t} \Delta w - \cancel{E_{1n} \frac{\Delta h}{2}} - \cancel{E_{2n} \frac{\Delta h}{2}} - E_{2t} \Delta w \\ &\quad + \cancel{E_{1n} \frac{\Delta h}{2}} + \cancel{E_{2n} \frac{\Delta h}{2}} \end{aligned}$$

$$\Rightarrow (E_{1t} - E_{2t}) \Delta w = 0$$

$\therefore \mathbf{E}_{1t} = \mathbf{E}_{2t} \rightarrow$ tangential components remain the same and undergo no changes

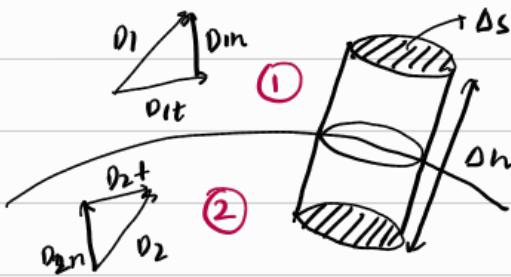
$$D = \epsilon E = D_t + D_n$$

$$\frac{D_{1t}}{\epsilon_1} = E_{1t} = E_{2t} = \frac{D_{2t}}{\epsilon_2} \Rightarrow \frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}$$

$$\frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2}$$

$\nearrow D_t$ undergoes same change across the interface

$D_n \rightarrow$ discontinuous across the surface



$\rightarrow$ contribution due to sides vanishes. ($\theta = 90^\circ$)

Allowing $\Delta h \rightarrow 0$

$$\Delta Q \approx \rho_s \Delta S \approx D_{1n} \Delta S - D_{2n} \Delta S$$

$$\rho_s = D_{1n} - D_{2n}$$

$\rho_s \rightarrow$ free charge density placed deliberately at the boundary

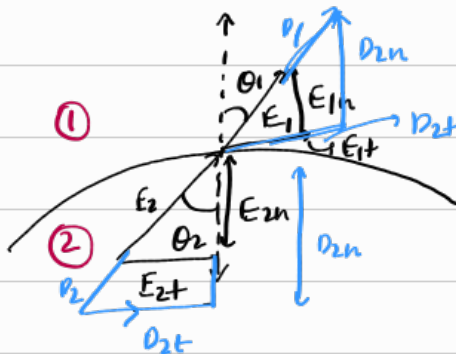
$\hat{n} \rightarrow$ directed from region 2 to region 1

$$\text{When } \rho_s = 0 \Rightarrow D_{1n} = D_{2n}$$

$$\Rightarrow \epsilon_1 E_{1n} = \epsilon_2 E_{2n}$$

$E_{1t} = E_{2t}$ and $D_{1n} = D_{2n}$ (or) $D_{1n} - D_{2n} = \rho_s$ are referred to as boundary conditions

$\rightarrow$ We can also find refraction of the Electric field across the interface.



$$E_1 \cos(90 - \theta_1) = E_{1t} = E_{2t} = E_2 \cos(90 - \theta_2)$$

$$E_1 \sin \theta_1 = E_{1t} = E_{2t} = E_2 \sin \theta_2$$

$$E_1 \sin \theta_1 = E_2 \sin \theta_2 \quad \text{--- (1)}$$

$$\epsilon_1 E_1 \cos \theta_1 = D_{1n} = D_{2n} = \epsilon_2 E_2 \cos \theta_2$$

$$\epsilon_1 E_1 \cos \theta_1 = \epsilon_2 E_2 \cos \theta_2 \quad \text{--- (2)}$$

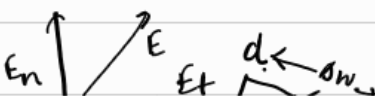
$$\frac{E_1 \sin \theta_1}{\epsilon_1 E_1 \cos \theta_1} = \frac{E_2 \sin \theta_2}{\epsilon_2 E_2 \cos \theta_2} \Rightarrow \frac{\tan \theta_1}{\epsilon_1} = \frac{\tan \theta_2}{\epsilon_2}$$

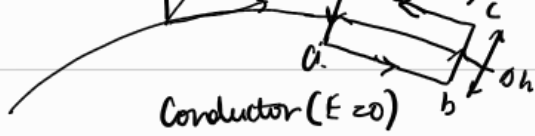
$$\Rightarrow \frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$$

(B) conductor-dielectric boundary condition:

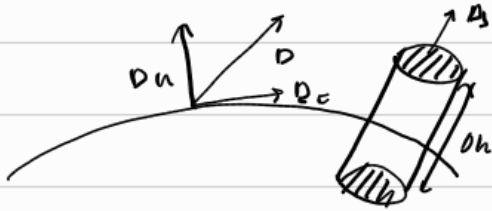
$\rightarrow$ conductor assumed to be perfect $\sigma \rightarrow \infty, \rho_c \rightarrow 0$

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0 = 0 \cdot \cancel{\Delta w} + 0 \cdot \frac{\Delta h}{2} + E_n \frac{\Delta h}{2} - E_t \Delta w - E_n \frac{\Delta h}{2} - 0 \cdot \cancel{\Delta h}$$





$$\text{As } \Delta h \rightarrow 0 \Rightarrow E_t \Delta h = 0 \Rightarrow \boxed{E_t = 0}$$



$$\text{As } \Delta h \rightarrow 0 \Rightarrow \Delta Q = D_n \Delta S - 0 \Delta S$$

$$\Rightarrow \rho_s \Rightarrow \frac{\Delta Q}{\Delta S} = D_n$$

$$(\text{or}) \quad D_n = \rho_s$$

→ following conclusions can be made abt a perfect conductor:

(1) no electric field may exist within a conductor

$$\therefore \rho_v = 0 \quad E = 0$$

(2) $E = -\nabla V = 0 \rightarrow$ no potential diff b/w any two points of the conductor

∴ conductor is an equipotential body

$$(3) \quad D_t = \epsilon_0 \cdot \epsilon_r \cdot E_t = 0 \quad D_n = \epsilon_0 \cdot \epsilon_r \cdot E_n = \rho_s$$